



Overview of Deep Learning

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github.com/roatienza

Updated: 2026

Artificial Intelligence (AI)

Machines exhibiting animal or human intelligence

Ants bridging algo

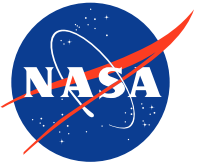


<https://www.quantamagazine.org/the-simple-algorithm-that-ants-use-to-build-bridges-20180226>



UP AI Research

ParSeq – SOTA STR (ECCV 2022)

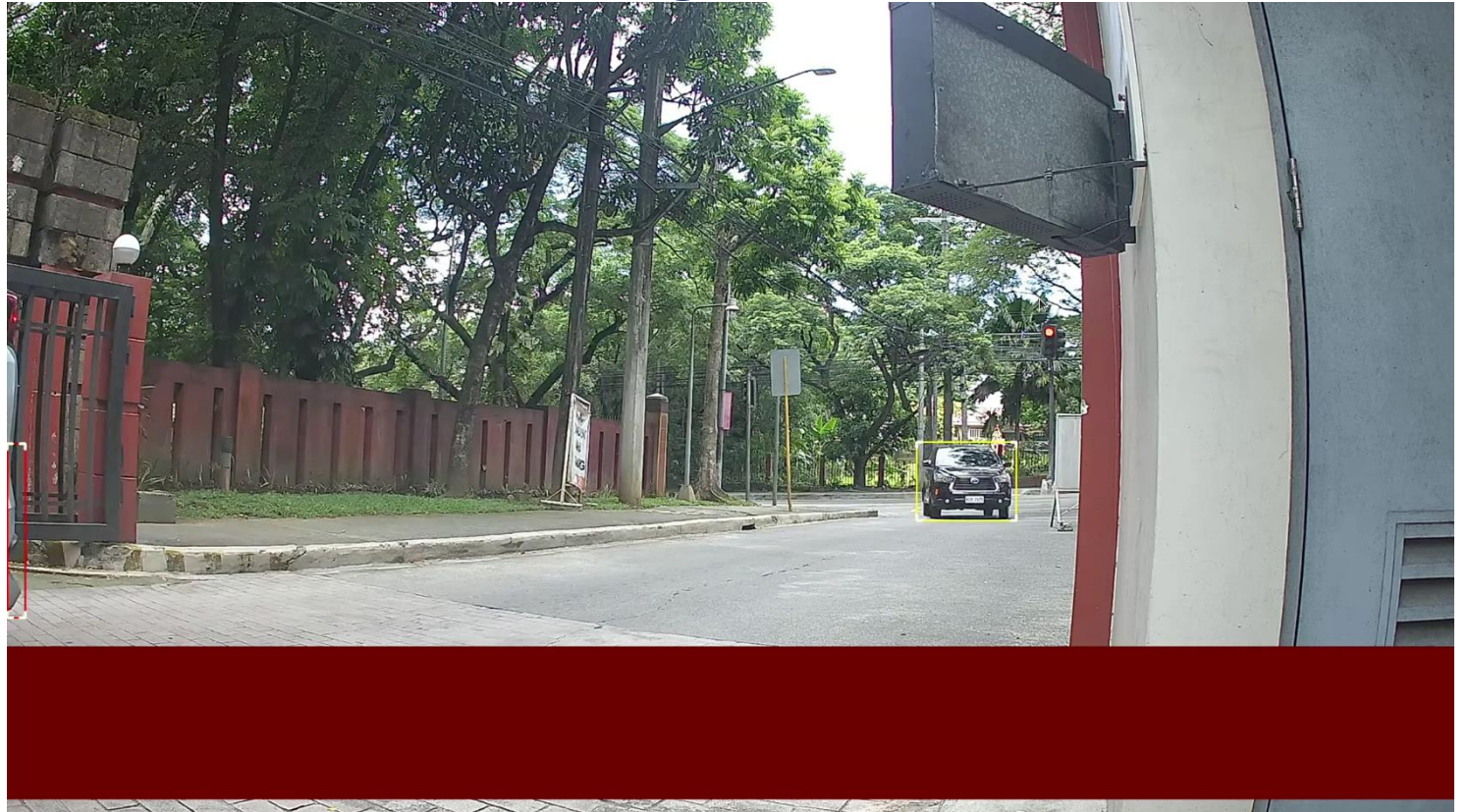


Used by NASA ISS



<https://www.nasa.gov/astrobee/>

Car Plate Recognition at UP Diliman



Bautista, Darwin, and Rowel Atienza. "Scene text recognition with permuted autoregressive sequence models." *European conference on computer vision*. Cham: Springer Nature Switzerland, 2022.

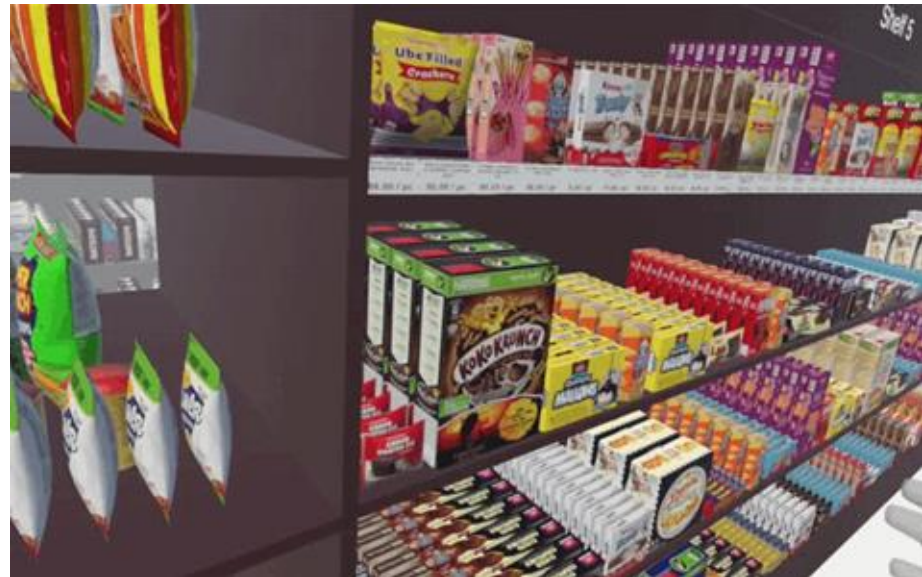
Sari Sandbox: A Virtual Retail Store Environment for Embodied AI Agents

To be presented: ICCV 2025 Workshops

Janika Deborah Gajo, Gerarld Paul Merales, Jerome E. Escarcha, Brenden Ashley Molina, Gian Nartea, Emmanuel Maminta, Juan Carlos Roldan, Rowel Atienza

<https://sarisandbox.github.io/>





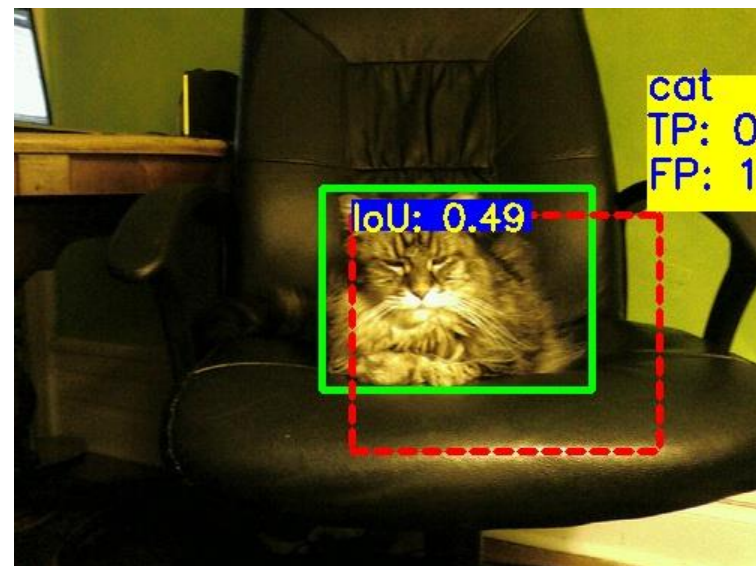
Interpretable Open-Vocabulary Referring Object Detection with Reverse Contrast Attention

To be presented: ICCV 2025 Workshops

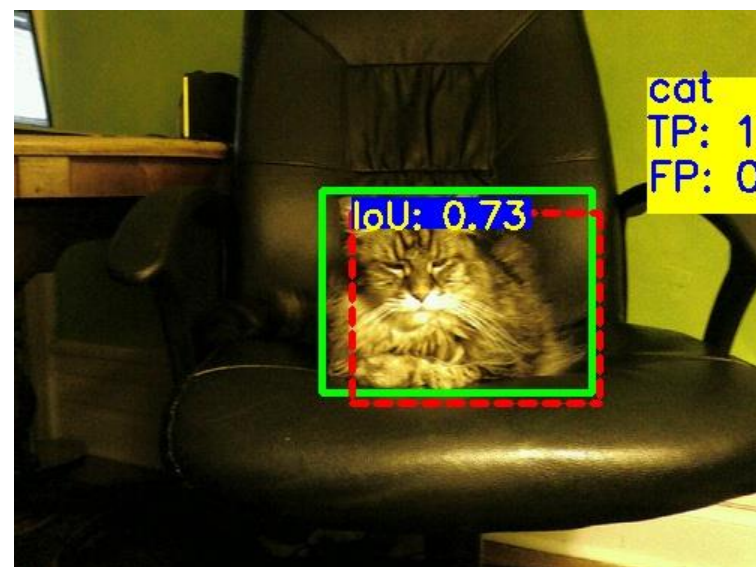
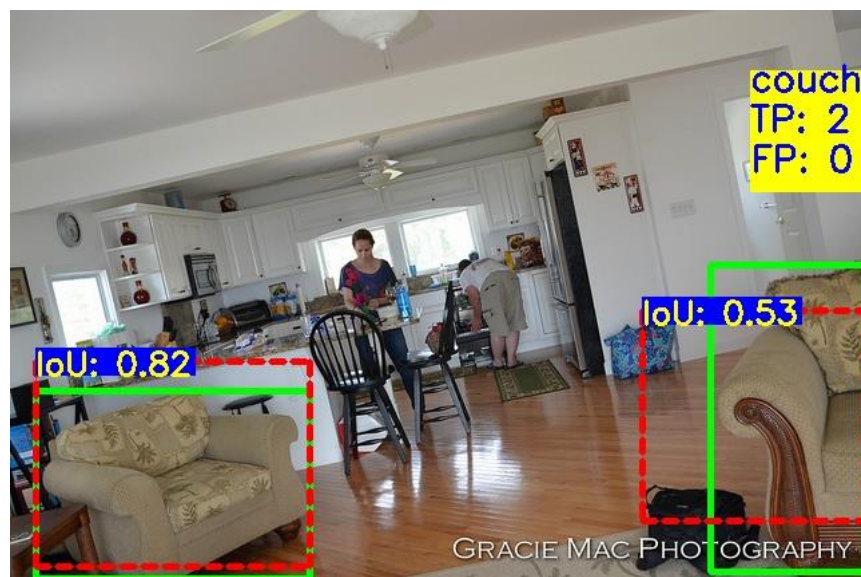
Drandreb Earl Juanico, Rowel Atienza and Jeffrey Go (SRPH)

Deepseek-VL2

Pre-RCA

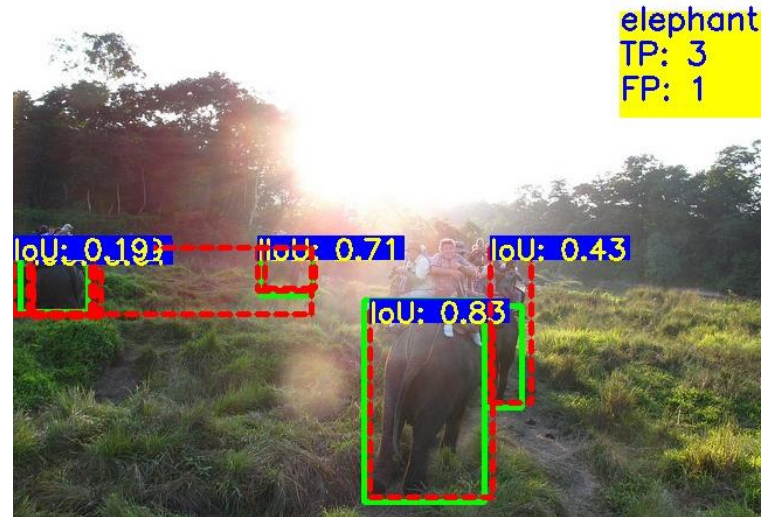


Post-RCA

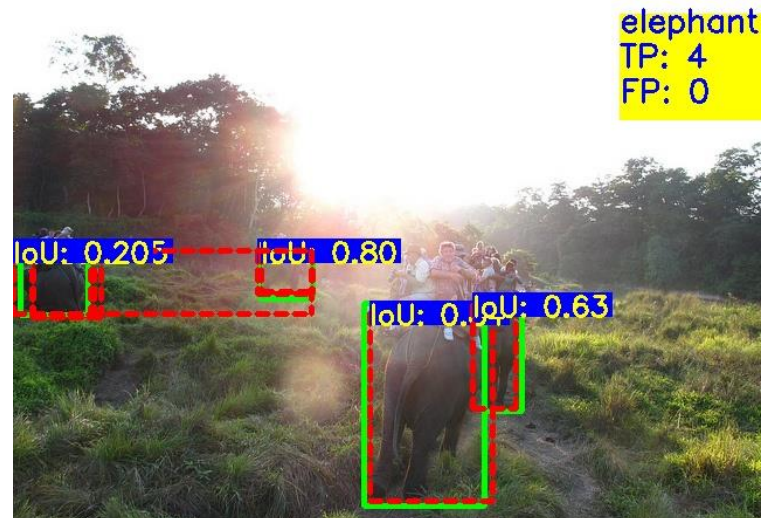


PaliGemma2

Pre-RCA



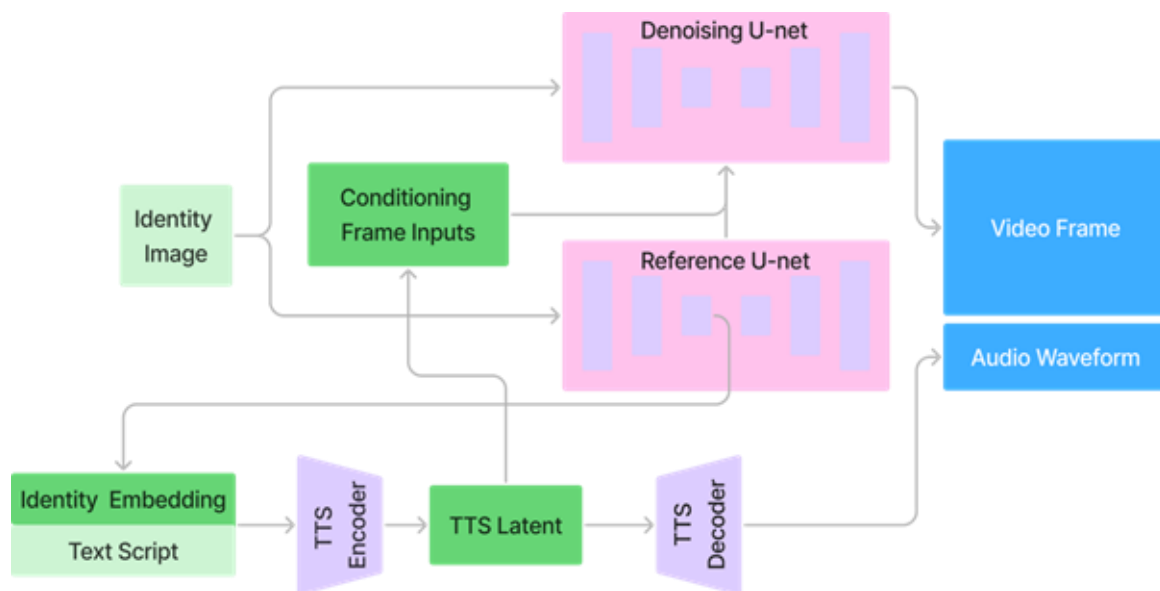
Post-RCA



Talking Head

Marcus Joseph Reyes, Rowel Atienza

Solution Overview



Adapt current visual and TTS models for our task to ensure good predictions with stable perspective

*Results are partial and preliminary

Partial Results Overview

Visual (Stable Perspective)



Audio-Visual Matching

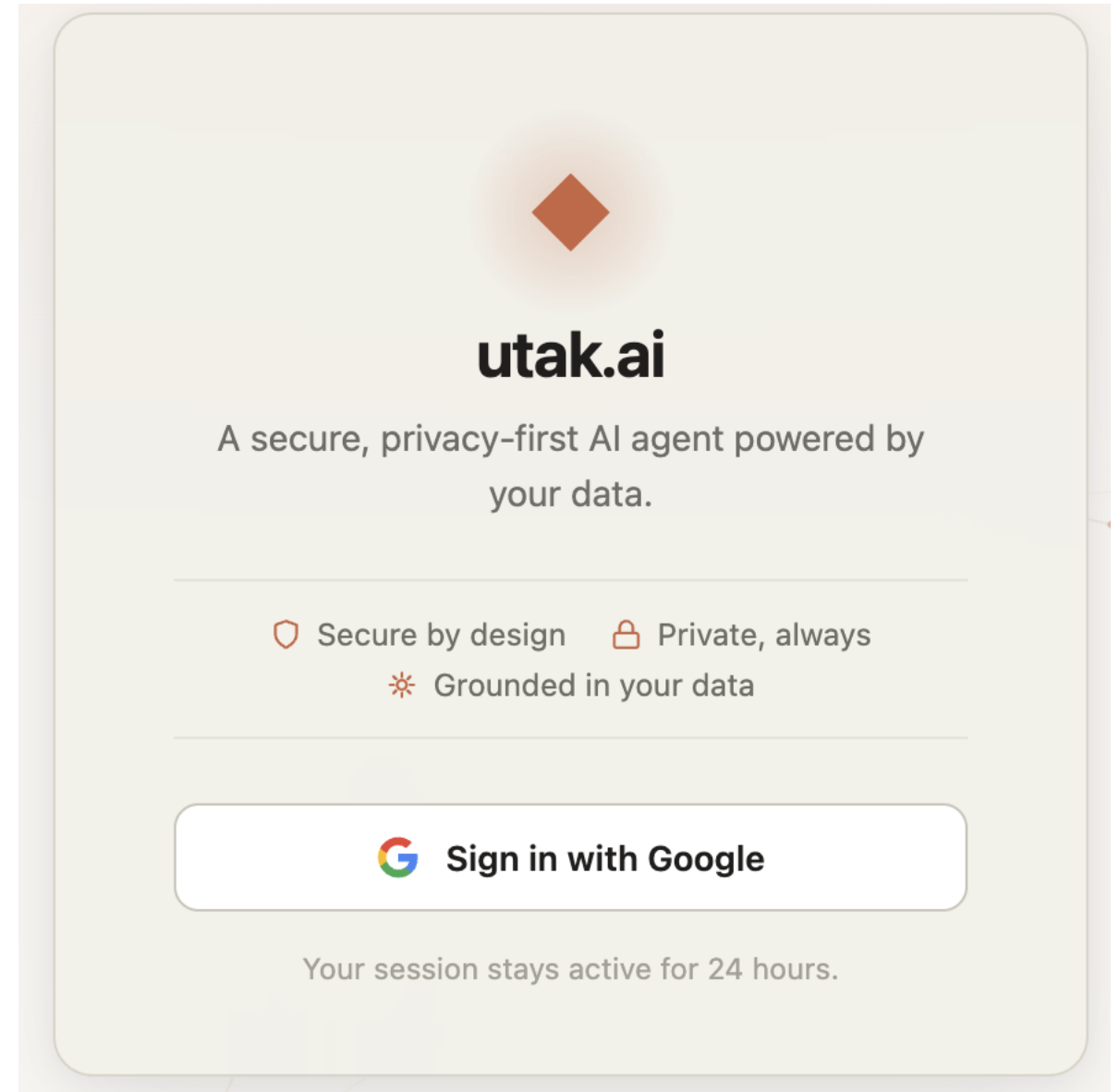


GT(left) vs Inferred (right)



On-it: Agent Harness

- <https://utak.ai>



Motivation – Use an army of SLMs instead of one super smart LLM

Small Language Models are the Future of Agentic AI

**Peter Belcak¹ Greg Heinrich¹ Shizhe Diao¹ Yonggan Fu¹ Xin Dong¹
Saurav Muralidharan¹ Yingyan Celine Lin^{1,2} Pavlo Molchanov¹
¹NVIDIA Research ²Georgia Institute of Technology
agents@nvidia.com**

Retal Vision



Rowel Atienza

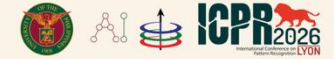
Deep Learning, University of th

WHAT MATTERS FOR GROCERY PRODUCT RETRIEVAL WITH OPEN-SOURCE VISION-LANGUAGE MODELS

Emmanuel G. Maminta¹ and Rowel O. Atienza^{1,2}

¹Artificial Intelligence Program, University of the Philippines

²Electrical and Electronics Engineering Institute, University of the Philippines



190

Open-source VLMs
Measured zero-shot

409

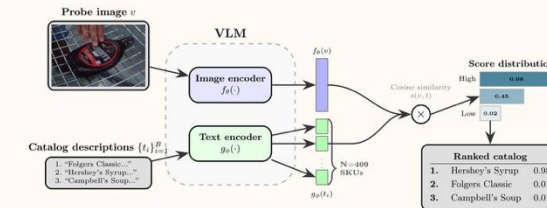
Grocery SKUs
One description each

74,200

In-situ shelf images
Real store conditions

WHICH VISION-LANGUAGE MODEL BELONGS IN A GROCERY STORE?

GROCERY PRODUCT RETRIEVAL



2025 GROCERYVISION CHALLENGE MPR TRACK

EASY



HARD



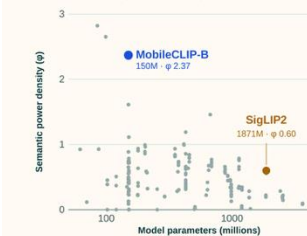
SEMANTIC POWER DENSITY

$$\phi = \frac{\left(\frac{\text{Recall@1}}{1 - \text{Recall@1} + \epsilon} \right)^2}{N_{\text{params}}} \times 100$$

$\epsilon = 10^{-6}$

THE BEST EDGE MODEL IS 12x SMALLER

MobileCLIP-B keeps 85% of the peak Recall@1



Each gray dot is one of the 190 checkpoints.
 ϕ is the signal for each million parameters.

THE LAST RANK IS THE HARD ONE

Best model, ViT-gopt-16-SigLIP2-384



The model is a good category retriever.
It is not a fine-grained discriminator.
A reranker can close this gap.

DATA BEATS SIZE

Change the data and not the model
Circle area shows the model size



A 150M model with clean data is more accurate than a 428M model with web data.

WHAT TO DEPLOY

EDGE

MobileCLIP-B
150M · 0.3 GB fp16
Jetson Orin Nano 8GB
Recall@1 0.683

STORE SERVER

PE-Core-L-14-336
671M · 1.3 GB fp16
NVIDIA L4 24GB
Recall@1 0.758

DATA CENTER

ViT-gopt-16-SigLIP2-384
1871M · 3.7 GB fp16
NVIDIA A100
Recall@1 0.770

Use 384px. A larger input costs 1.7 to 2.2 times more compute and does not increase Recall@1. The hardware shows a memory budget class. It is not a tested platform.



Code and all 190 results



A Survey of Spatial Memory Representations for Efficient Robot Navigation

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University of the Philippines Diliman¹, University of Santo Tomas²

CVPR
JUNE 3-7, 2026



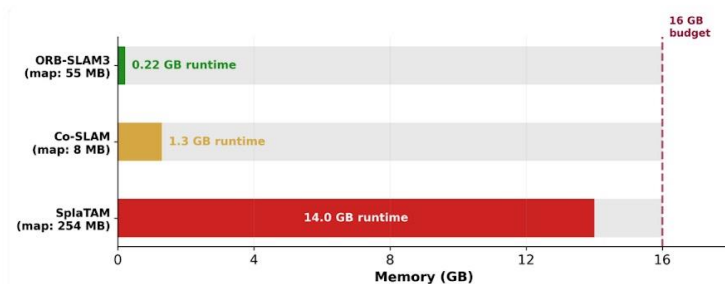
Why memory efficiency matters

- Robotic navigation requires **persistent spatial representations** for localization and mapping.
- **Memory grows unboundedly** with environment size and mission duration.
- Embedded platforms (8–16 GB, <30 W) cannot be upgraded at deployment.

The Consequences Compound



The Memory Bottleneck



Map size alone does not predict deployment feasibility.

SplatAM consumes 87%, leaving less than 2 GB for perception, planning, and the OS. Co-SLAM's 8 MB map looks safe, until runtime reveals 1.3 GB actual cost.

The Overhead Factor, α

Published map sizes are misleading proxies for deployment cost. To capture this gap, we define the overhead factor:

$$\alpha = \frac{M_{peak}}{M_{map}}$$

where:

- M_{peak} is the **peak runtime memory**
- M_{map} is the **total persistent checkpoint size on disk**

Note: **α CPU** (CPU RSS, used for sparse systems) while **α GPU** (GPU device allocation, used for neural systems).

What does it mean?

Low $\downarrow$ map size predicts deployment cost

High $\uparrow$ hidden runtime expenses

Example:

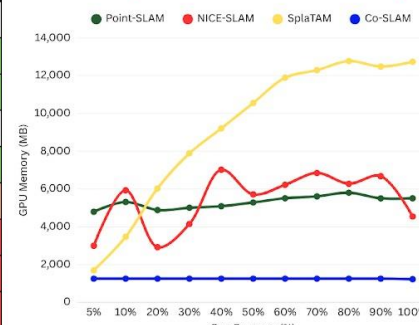
Co-SLAM saves an 8 MB map to disk ($M_{map} = 8$ MB) but requires 1.3 GB of GPU memory at runtime ($M_{peak} = 1,258$ MB)

$$\alpha = \frac{1,258}{8} = 157 \rightarrow \text{The map looks small. The deployment cost is not.}$$

Profiling reveals 100x α variation

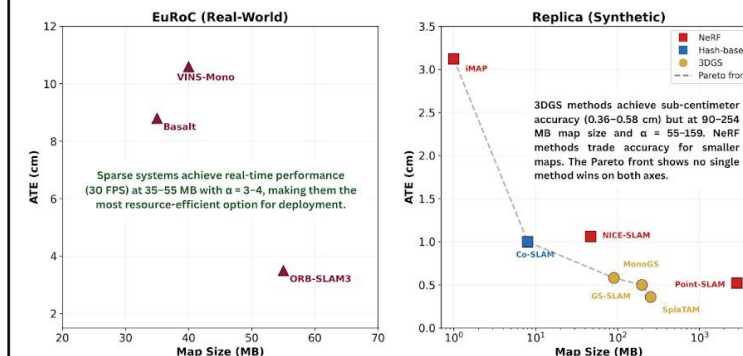
System	ATE	Map	Peak	FPS	α
Sparse - EuRoC (real-world)					
ORB-SLAM3	3.5cm	55MB	220MB	30fps	4.0
Basalt	8.8cm	35MB	120MB	30fps	3.4
Neural - Replica (synthetic)					
Point-SLAM	0.52cm	2865MB	6563MB	1fps	2.3
SplatAM	0.36cm	254MB	14024MB	-	55
Co-SLAM	1.00cm	8MB	1258MB	16fps	157
SGS-SLAM	0.41cm	254MB	40330MB	2fps	159
NICE-SLAM	1.06cm	47MB	10082MB	2fps	215

Memory Grows Differently



Low α does not mean low cost; high α hides runtime expenses invisible in static map sizes.

No Single Paradigm Dominates



Sparse $\rightarrow$ Most Efficient; 3DGS $\rightarrow$ Most Accurate.

How large a map can my hardware support?

$$M_{map_{max}} = \frac{M_{budget}}{\alpha}$$

where:

- $M_{map_{max}}$ is the **maximum feasible map size**
- M_{budget} is the **available memory after OS and drivers**
- α is the **overhead factor**

Representation Selection Guide

Constraint	Recommended	α range	Type	Max Map	Avoid
CPU-only (<8 GB)	Sparse, Octree	3 - 5	CPU	~4 GB	Neural
Embedded GPU (<16 GB)	Sparse, SG	4 - 10	CPU	~2 GB	Raw 3DGS
Dense Geometry	TSDF, 3DGS	~55	GPU	290 MB	Sparse only
Photo rendering	3DGS, NeRF	2 - 215	GPU	75 MB	Occupancy
Multi-hour	Submaps, Stream	varies	-	-	Monolithic
Semantic	SG, VLM	4 - 10	CPU	-	Geom. only



<https://arxiv.org/abs/2604.16482>

<https://spatial-memory.github.io>

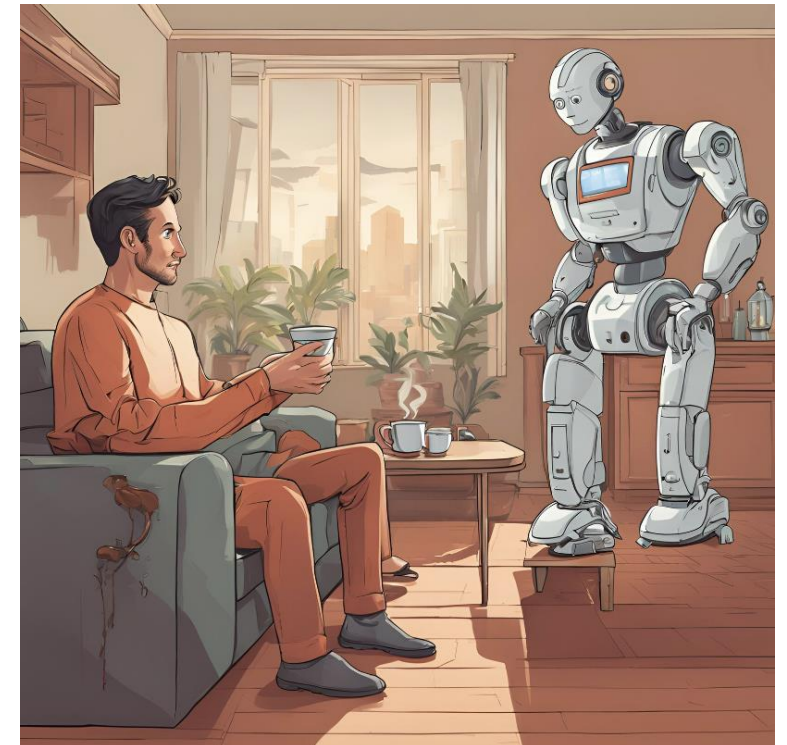


Intelligence

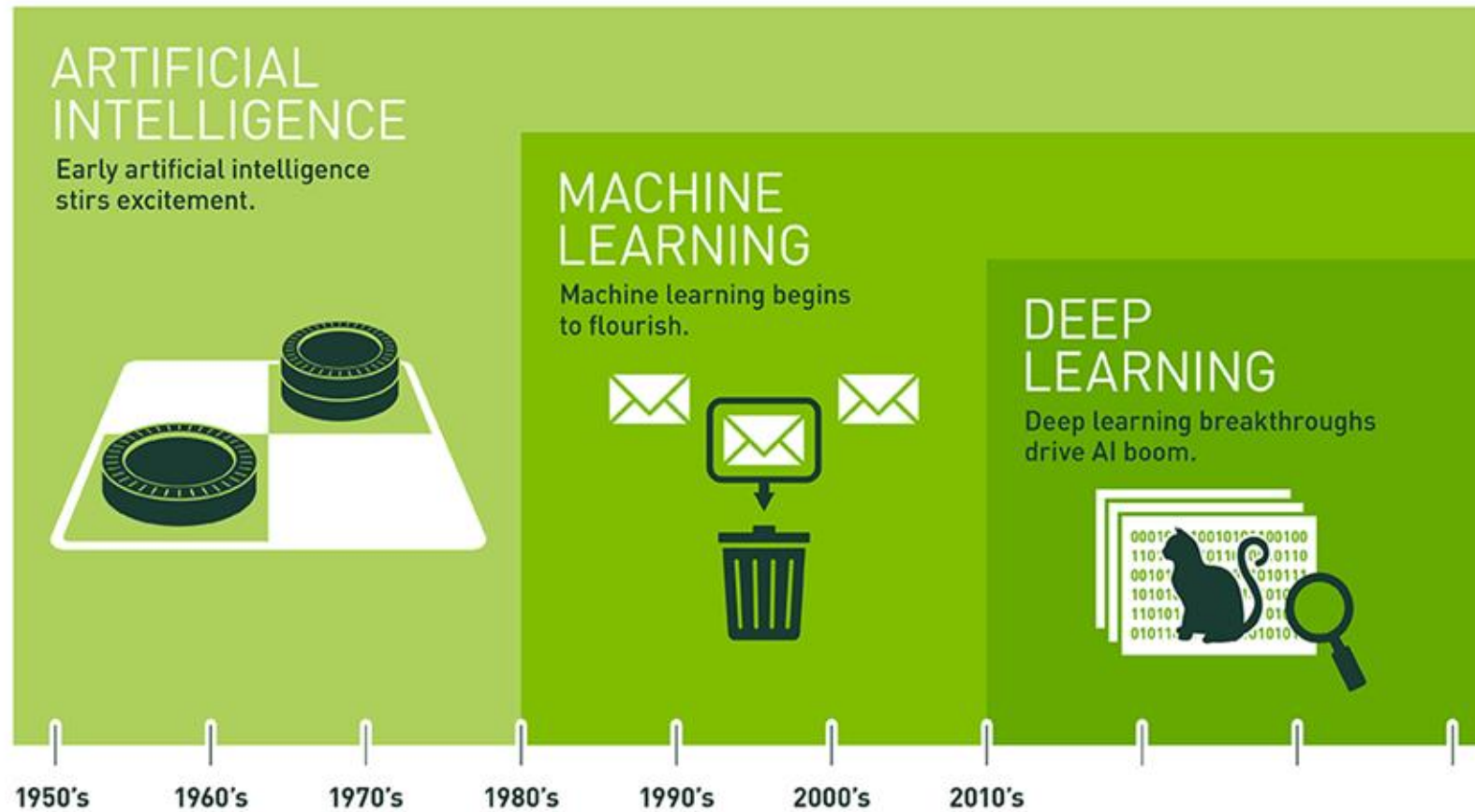
- A very general mental capability that among other things involves the ability to:
 - Reason
 - Plan
 - Solve problems
 - Think abstractly
 - Comprehend complex ideas
 - Learning quickly and learning from experience

Journal of Intelligence 1997 Vol 24 No 1

Get me a
cup of
coffee.



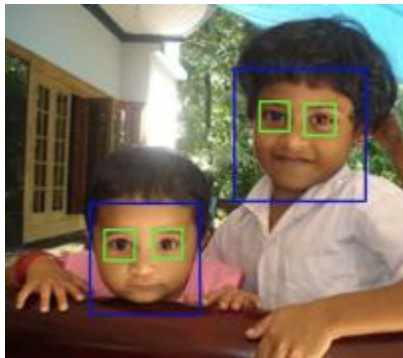
AI, Machine Learning and Deep Learning



NVIDIA

Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

Artificial Intelligence, Machine Learning and Deep Learning on Face Detection



Rule-based (AI):

Detect facial features based on color/template

Apply if-else-if-else

[Viola & Jones 2001]



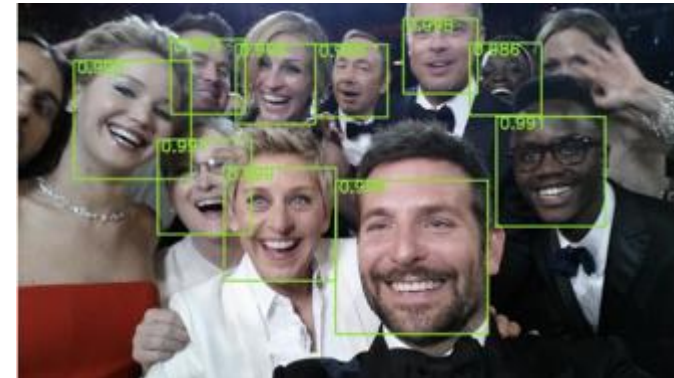
Machine Learning:

Use Haar Cascade

Classifier

Hand-crafted feature detection

YOLO, SSD, RCNN [>2012]



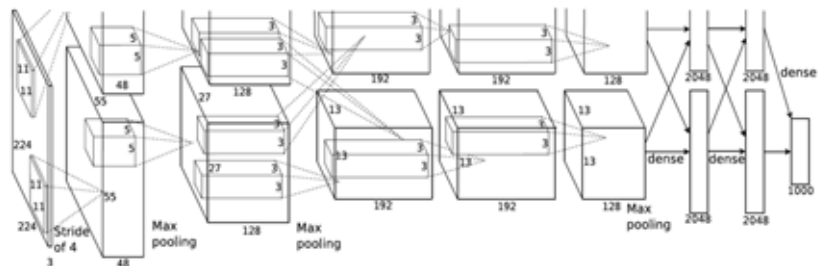
Deep Learning:

Train a network by showing thousands of labelled region of faces

Automatic feature detection

Deep Learning Moment - 2012

- AlexNet – 650,000-neuron deep neural network won 2012 ImageNet1k competition with 15% Top-5 test error rate **compared** to 2nd place with 26%. Its top 1 val accuracy is 63.3%.



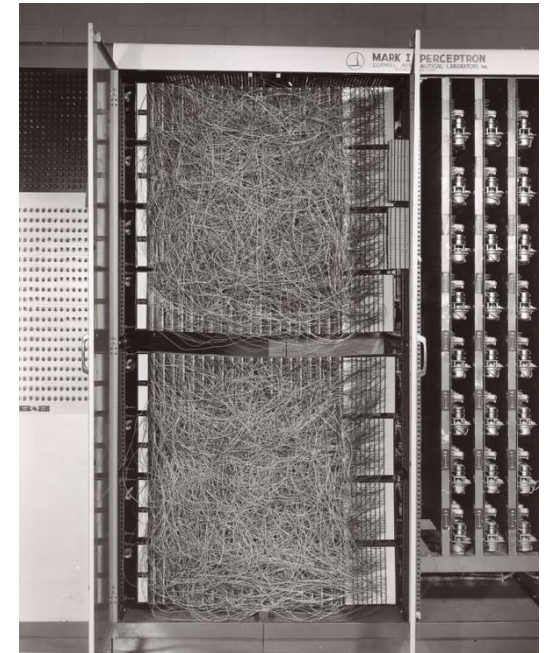
Model	Top-1 (val)	Top-5 (val)	Top-5 (test)
<i>SIFT + FVs [7]</i>	—	—	26.2%
1 CNN	40.7%	18.2%	—
5 CNNs	38.1%	16.4%	16.4%
1 CNN*	39.0%	16.6%	—
7 CNNs*	36.7%	15.4%	15.3%

Table 2: Comparison of error rates on ILSVRC-2012 validation and test sets. In *italics* are best results achieved by others. Models with an asterisk* were “pre-trained” to classify the entire ImageNet 2011 Fall release. See Section 6 for details.

A. Krizhevsky, I. Sutskever, G. Hinton, [ImageNet Classification with Deep Convolutional Neural Networks](#) (2012), NeurIPS

Deep Learning - what made it work?

- Concepts of artificial neural network (ANN) and convolutional neural network (CNN) are old
 - Neurons in perceptron (1-layer NN) – 1958
 - Neocognitron (1980) and CNN (1989)
 - Backpropagation (1986)
- What's new?
 - Computing power – Massive number of GPU CUDA cores
 - Data – from the Internet

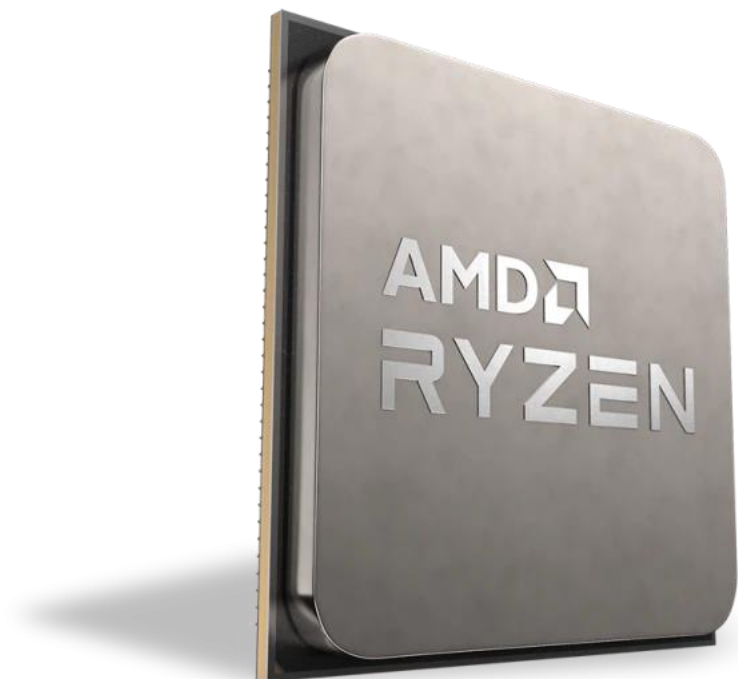


Perceptron is a binary classifier

CPU (AMD RyZen)

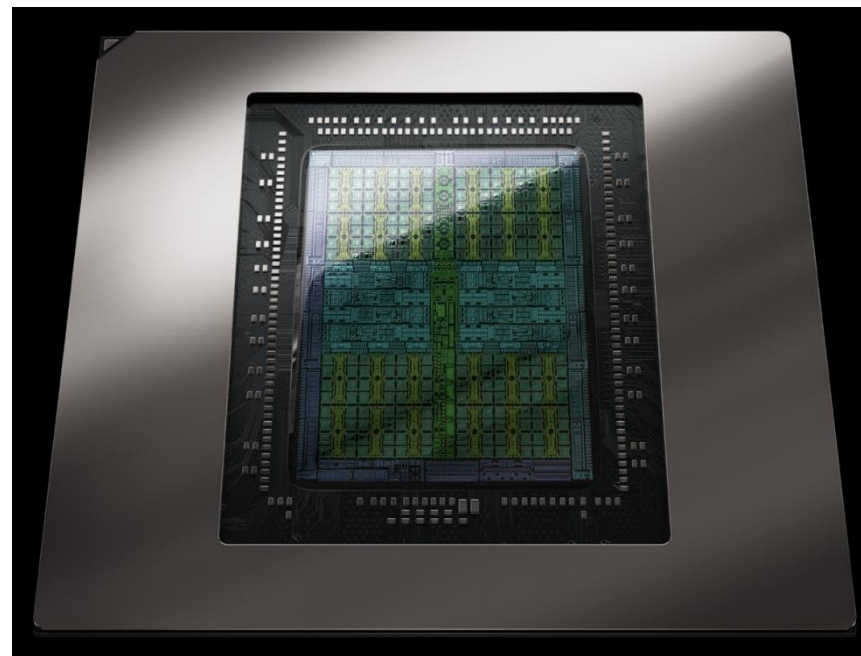
vs

GPU (RTX 5090)



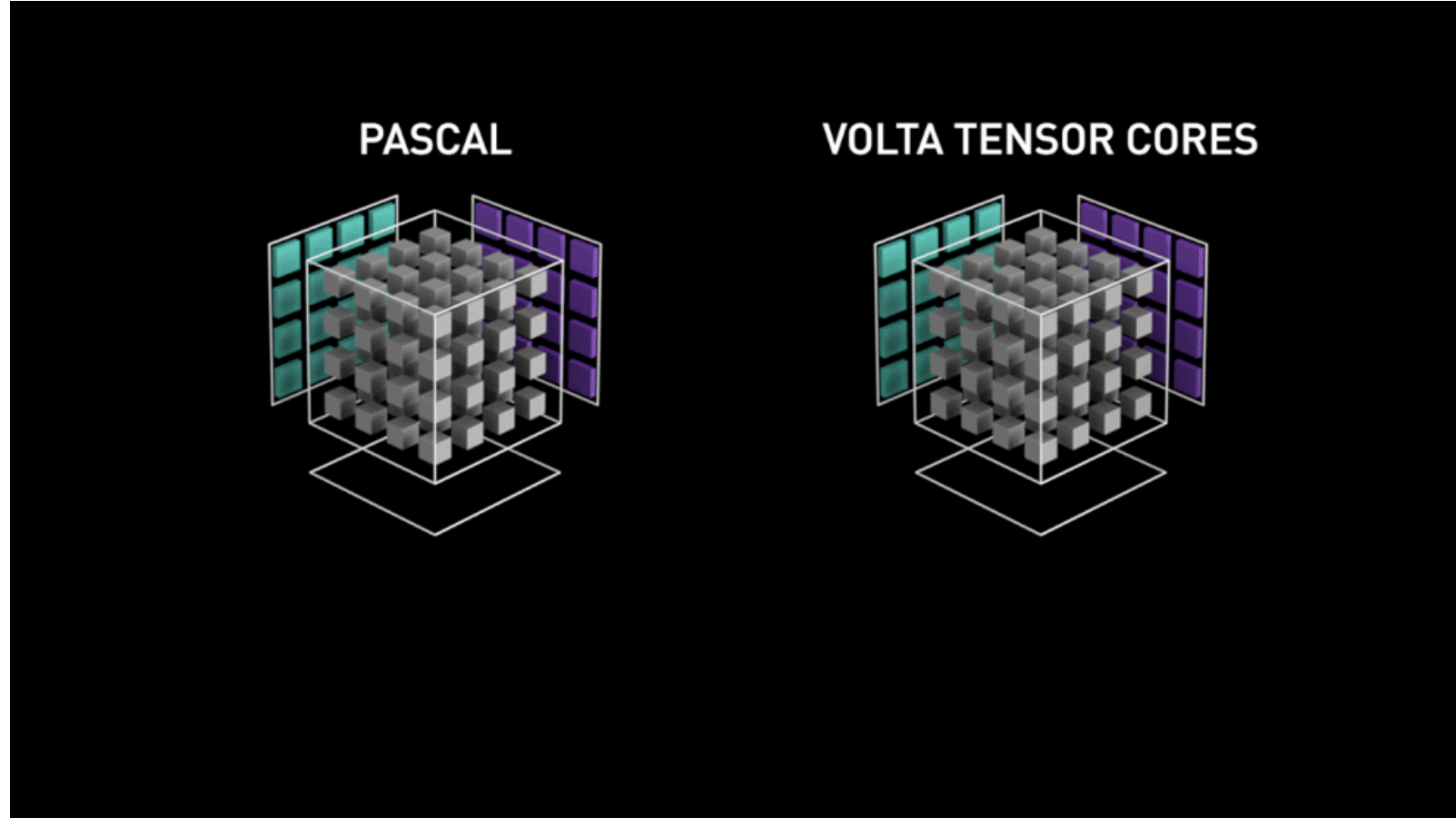
Ryzen 9 9950X3D, 16 cores and 32 threads, boost clock of up to 5.7 GHz.

Rowel Atienza

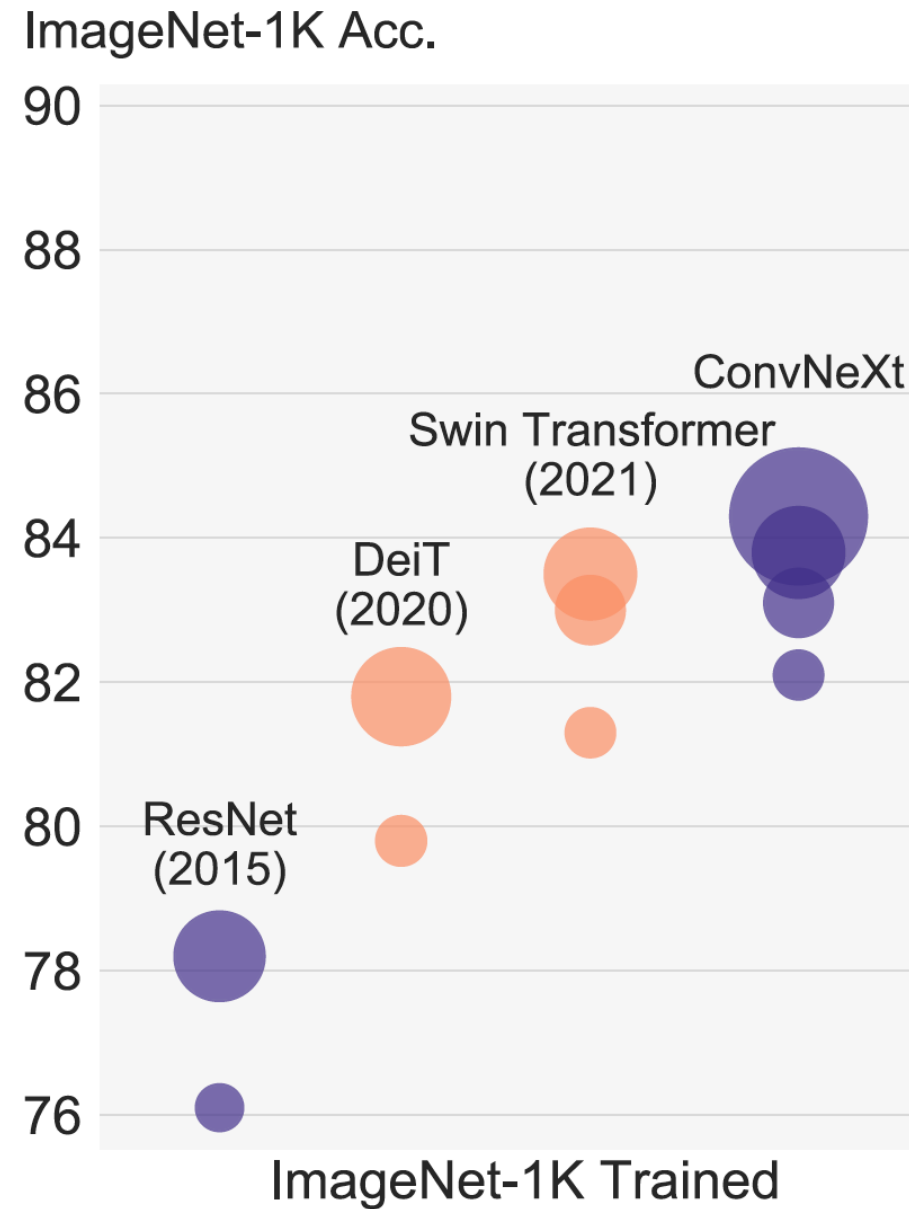


21,760 CUDA cores, 32GB of GDDR7 memory with a 512-bit bus, and a boost clock of up to 2.41 GHz

1D vs 2D Tensor Operation

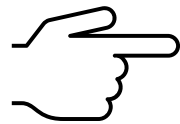


The Rest is History



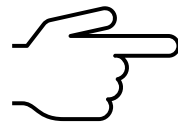
Barely scratching the surface of Artificial General Intelligence (AGI)

Starting to move here



Doing : $p(y|do(x))$

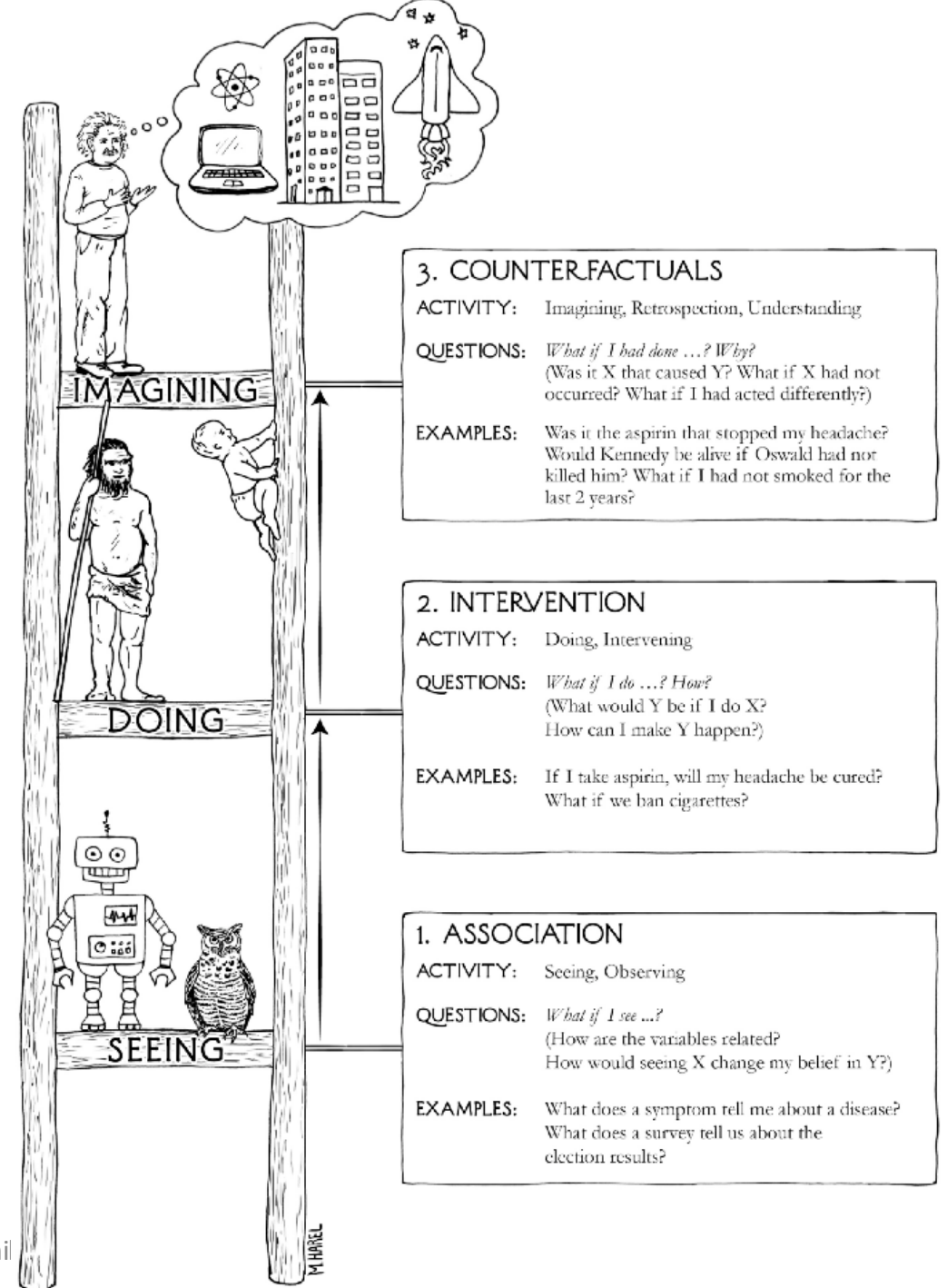
We are here



Seeing: $p(y|x)$

Pearl, Book of Why

Imagining



A robot thinking about crossing the river

Imagining

Doing : $p(\mathbf{y}|\text{do}(\mathbf{x}))$

Seeing: $p(\mathbf{y}|\mathbf{x})$



Do foundation models provide the path to AGI?

Takeaway question

References

Strickland, E. The Turbulent Past and Uncertain Future of Artificial Intelligencels there a way out of AI's boom-and-bust cycle?

<https://spectrum-ieee-org.cdn.ampproject.org/c/s/spectrum.ieee.org/amp/history-of-ai-2655064200>

Liu, Zhuang, et al. "A ConvNet for the 2020s." arXiv preprint arXiv:2201.03545 (2022).

NVIDIA.com

AMD.com

Dziri, Nouha, et al. "Faith and Fate: Limits of Transformers on Compositionality." *arXiv preprint arXiv:2305.18654* (2023).

End